深度學習
Deep Learning

Recurrent Neural Networks (RNN)

Instructor: 林英嘉 (Ying-Jia Lin)
2025/10/22

Course GitHub

Slido # DL1022

Outline

- Introduction to Natural Language Processing (NLP) [30 min]
- Recurrent Neural Networks (RNN) [50 min]
- Homework 2 [50 min]
- Quiz [20 min]

The Revolution of ChatGPT

ChatGPT came out in November, 2022.

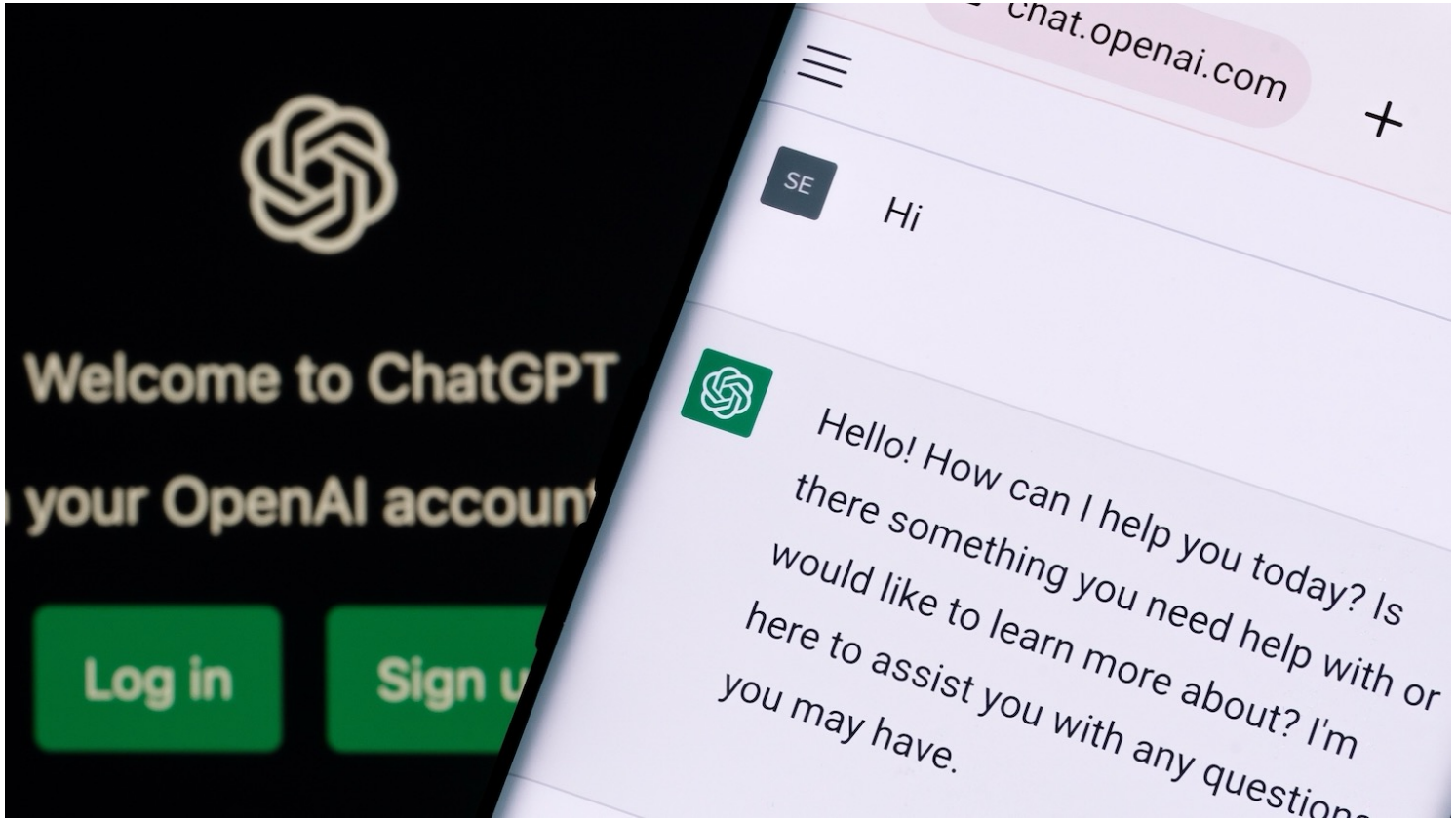


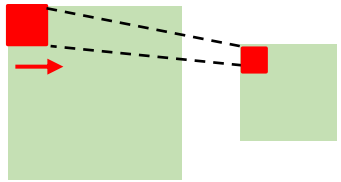
Figure source: https://poole.ncsu.edu/thought-leadership/article/lets-chat-about-chatgpt/

大型語言模型的應用

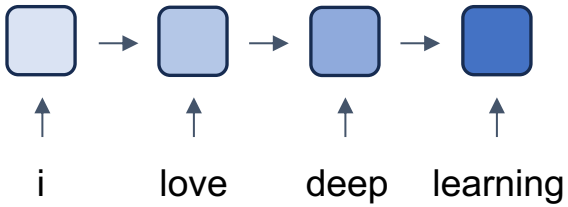
翻譯、聊天機器人、寫故事、
摘要、教學、
寫程式、寫作業 ...

深度學習常見模型架構

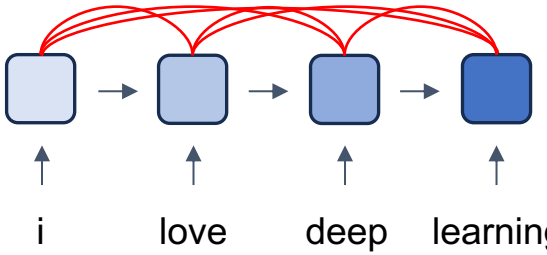
CNN
(著重局部資訊)



RNN
(著重記憶力)



Transformer
(著重全局資訊與記憶力)



Natural Language Processing (NLP)

電腦是如何學習自然語言呢？

NLP 三大秘辛

1. 每個字，都是一段向量
2. 每個 NLP 模型都有一個字典 
3. 自然語言處理模型需要記憶力 🧠

影像 vs. 文字

⟶ 28 x 28 的矩陣

Apple ⟶ ?

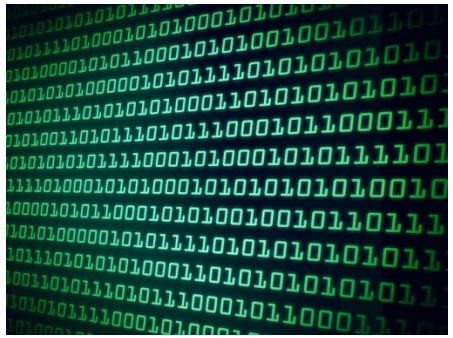


Figure source: https://kopu.chat/wp-content/uploads/2017/04/binary-507x380.jpeg

每個字，都是一段向量

可以自行設定向量長度

she = [0.110960 0.032115 -0.004809]

her = [0.110950 0.032117 -0.004909]

grandpa = [0.023115 0.708091, 0.512345]

grandma = [0.023113 0.708101, 0.512445]

Word Representations

like

apple

?

Basic Approach

Corpus

I like to eat apple

You like to watch TV

Vocabulary

apple	[1, 0, 0, 0, 0, 0, 0, 0]
eat	[0, 1, 0, 0, 0, 0, 0, 0]
I	[0, 0, 1, 0, 0, 0, 0, 0]
like	[0, 0, 0, 1, 0, 0, 0, 0]
watch	[0, 0, 0, 0, 1, 0, 0, 0]
to	[0, 0, 0, 0, 0, 1, 0, 0]
⋮	

One hot encodings with each vector size equal to the vocab size (8 in this case)

Pros: handy, training-free
Cons: sparse matrix which occupies much memory

nn.Embeddings

```python
word_to_idx = {"好吃": 0, "棒": 1, "给力": 2}
embeds = torch.nn.Embedding(3, 5)  # 2 words in vocab, 5 dimensional embeddings
```

```python
lookup_tensor = torch.tensor(
    [
        word_to_idx["好吃"],
        word_to_idx["棒"],
        word_to_idx["给力"],
    ],
    dtype=torch.long,
)
word_embed = embeds(lookup_tensor)
print(word_embed)
```

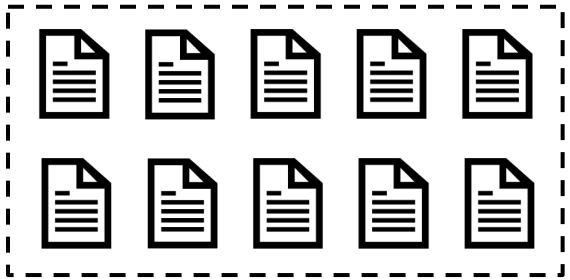
[7]

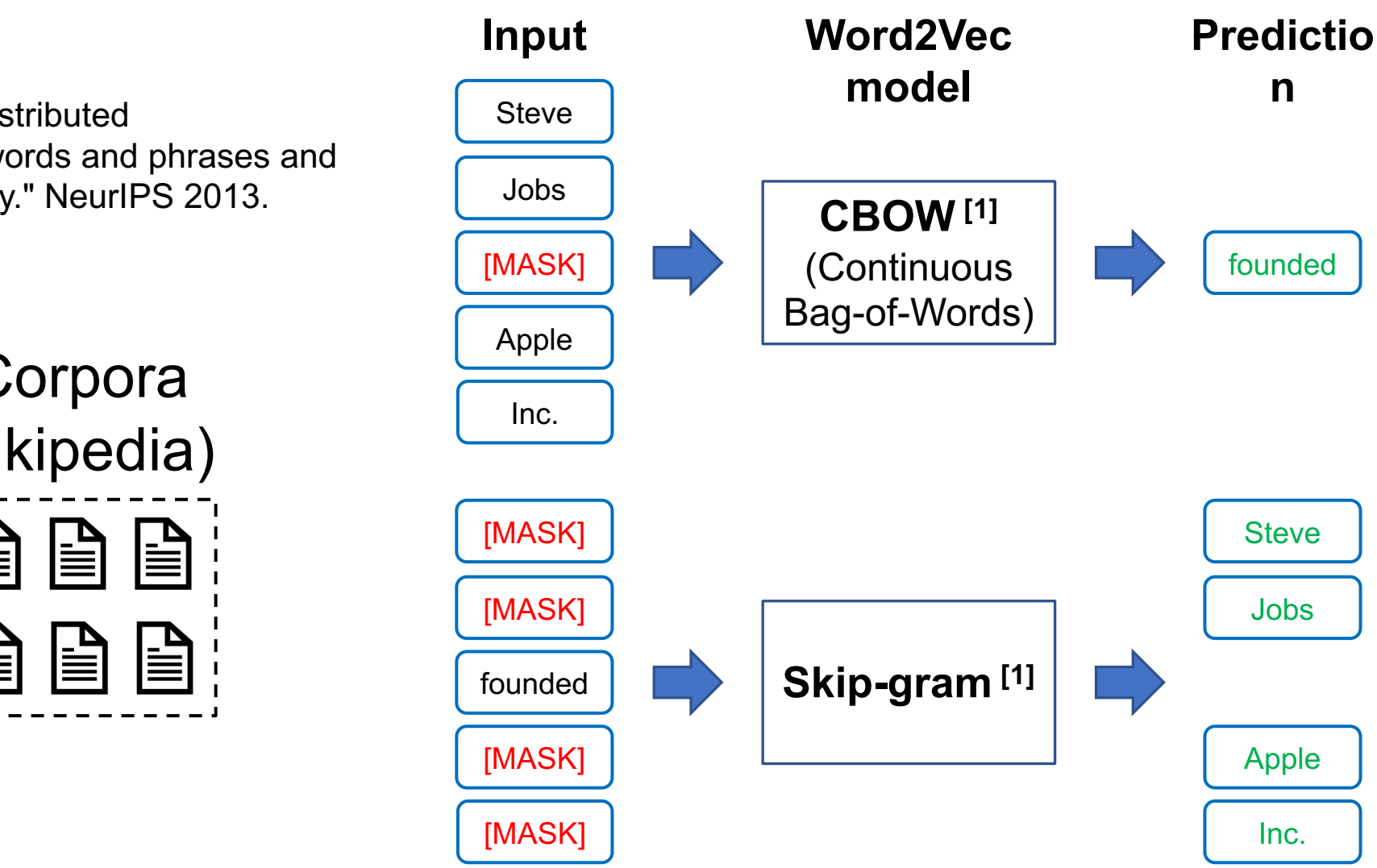
```
tensor([[ 2.2014,  1.9880,  1.5726, -0.1552,  0.0475],
        [-0.0540, -1.5926, -0.5808, -0.6631, -0.3321],
        [-0.3562,  1.0019,  0.8178, -0.9174, -0.0707]],
       grad_fn=<EmbeddingBackward0>)
```

Word Embeddings (Word2Vec)

[1] Mikolov et al. "Distributed representations of words and phrases and their compositionality." NeurIPS 2013.

Large Corpora (E.g., Wikipedia)





After Training Word Embeddings

- The outputs are dense vectors with a fixed dimension size:

Dimension size (e.g., 300)

Vocabulary size (e.g., 30,000)

apple -0.110960 0.016115 -0.004809 0.033589 0.121455 …
banana -0.027713 -0.015676 0.003314 0.077602 0.159718 …
…
…
…

NLP 三大秘辛

1. 每個字，都是一段向量
2. 每個 NLP 模型都有一個字典 
3. 自然語言處理模型需要記憶力 🧠

每個 NLP 模型都有一個字典 

Vocabulary

apple
eat
I
like
watch
to
⋮

→ LM → I like to eat apple watch

字典裡面的內容是語言模型能理解的最小單元，
如此一來模型才能夠進行目標任務 (e.g., 生成)

→  → ... (你平常操作 ChatGPT 的結果)

字典實作

字典的變數
(key: 字, value: index)

通常我們會預先準備 <pad> 跟 <unk>
前者代表 padding，後者代表 unknown words

```python
vocab = {'<pad>':0, '<unk>':1}

# 使字詞不重複
words = sorted(set(words))
for idx, word in enumerate(words):
    # 一開始已經放兩個進去 dictionary 了
    idx = idx + 2
    # 將 word to id 放到 dictionary
    vocab[word] = idx
```

Python

字典在NLP中的實作方式就真的採用Python dictionary

[注意事項] 字典實作

字典是經由訓練資料集的字所建立的

```
word_to_idx = {"<pad>": 0, "<unk>": 1, "好吃": 2, "棒": 3, "给力": 4}
embeds = torch.nn.Embedding(5, 5)  # 5 words in vocab, 5 dimensional embeddings
```

```
def get_word_id(word, vocab, unk_idx: int = 1):
    return vocab.get(word, unk_idx)
```

如果某字不存在於字典中，則通常給予 <unk> 的 id

```
lookup_tensor = torch.tensor(
    [
        get_word_id("好吃", word_to_idx),
        get_word_id("不棒", word_to_idx),
        get_word_id("<unk>", word_to_idx),
    ],
)
word_embed = embeds(lookup_tensor)
print(word_embed)
```

```
tensor([[-0.5574,  0.1768,  0.9588,  1.2837, -1.7256],
        [-3.4725, -1.0435, -0.0538, -0.0625, -0.4173],
        [-3.4725, -1.0435, -0.0538, -0.0625, -0.4173]],
       grad_fn=<EmbeddingBackward0>)
```

NLP 三大秘辛

1. 每個字，都是一段向量
2. 每個 NLP 模型都有一個字典 
3. 自然語言處理模型需要記憶力 🧠

語言的模式

⭐ 上下文非常重要

句子1：

小明昨天遇到小美，他對她說：「下次一起去看電影吧！」，請問小明下次想去跟誰看電影？

句子2：

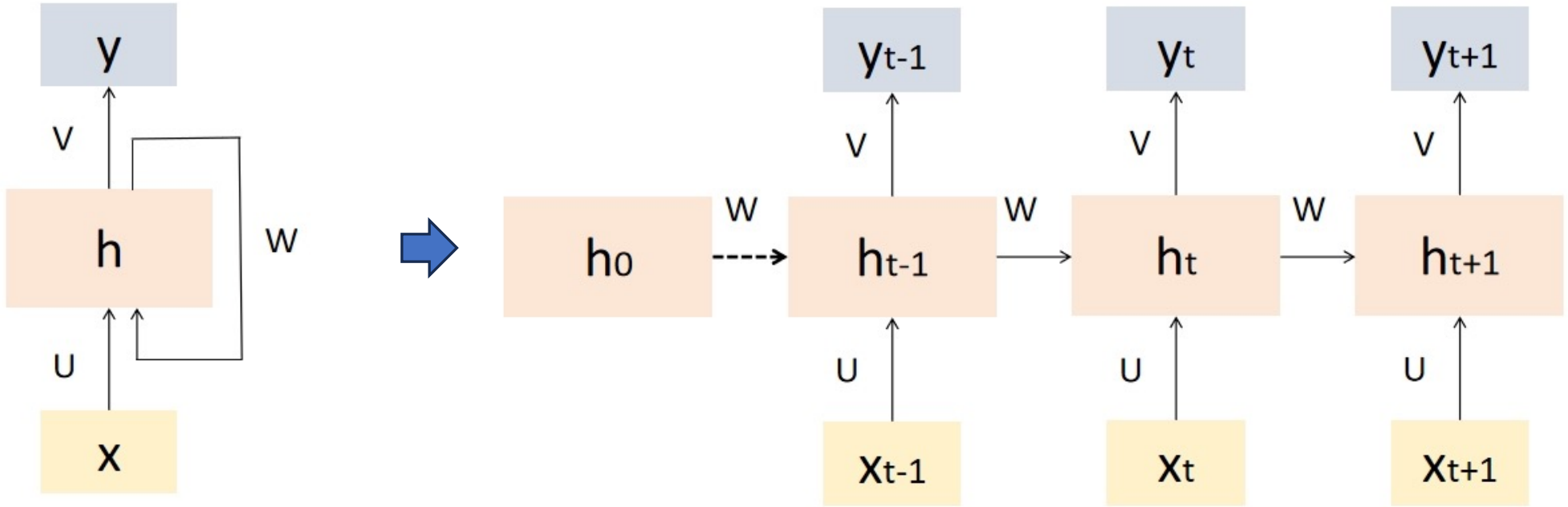
「庭院深深深幾許」，請問有幾個深？

Recurrent Neural Networks

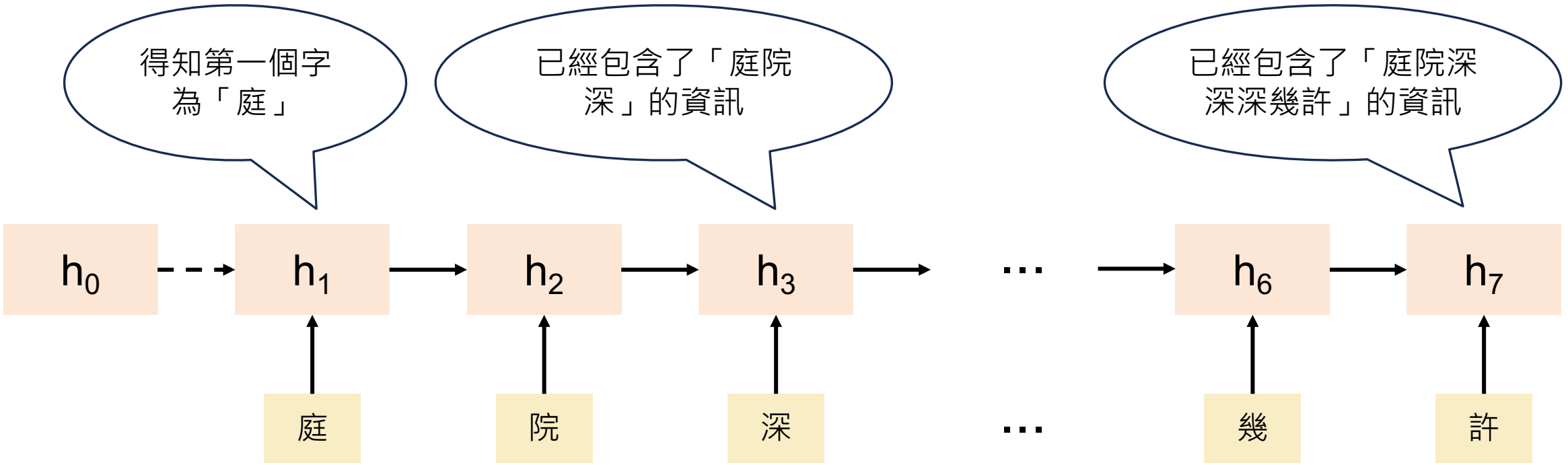
RNN

Recurrent Neural Networks (RNN)

- Recurrent Neural Networks (RNNs) are a type of artificial neural network designed to handle sequential data by capturing temporal dependencies.
  - RNN 的“遞迴”是指它在每一個時間步中重複使用相同的參數（同一組權重），並把先前的隱藏狀態傳到下一個時間步



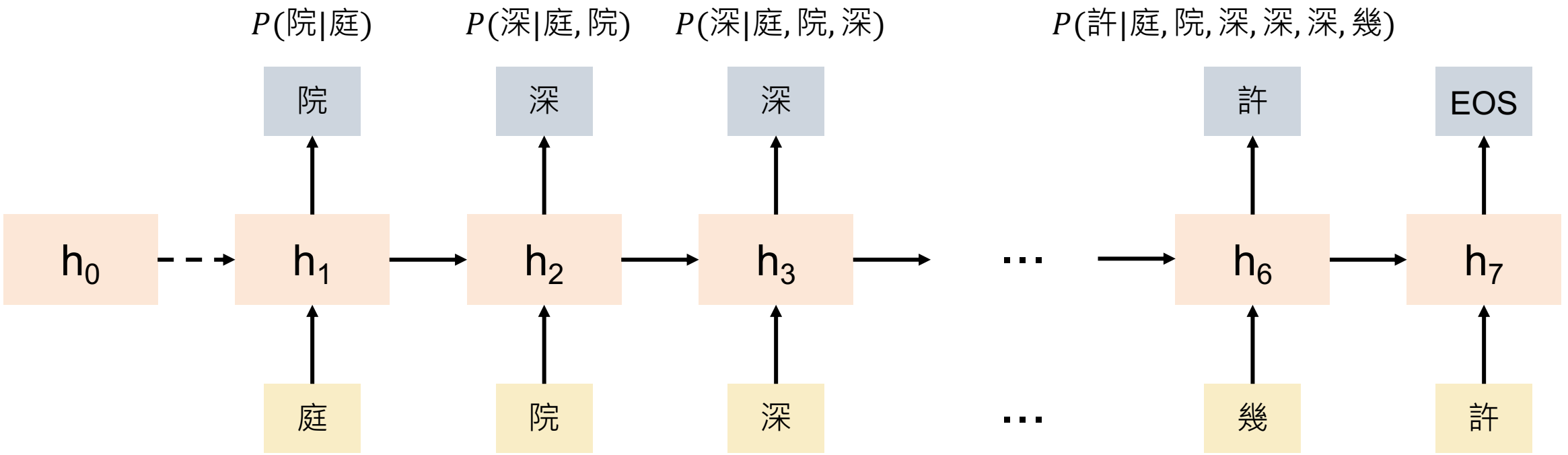
RNN 的記憶力意義


得知第一個字為「庭」
已經包含了「庭院深」的資訊
已經包含了「庭院深深深幾許」的資訊
h_0
h_1
h_2
h_3
…
h_6
h_7
庭
院
深
…
幾
許

RNN 以文字接龍進行訓練

EOS: End of Sentence

$P(\text{EOS}|\text{庭, 院, 深, 深, 深, 幾, 許})$



目標函數：log($P(\text{院}|\text{庭})\times P(\text{深}|\text{庭, 院})\times P(\text{深}|\text{庭, 院, 深})\times \ldots \times P(\text{EOS}|\text{庭, 院, 深, 深, 深, 幾, 許})$)

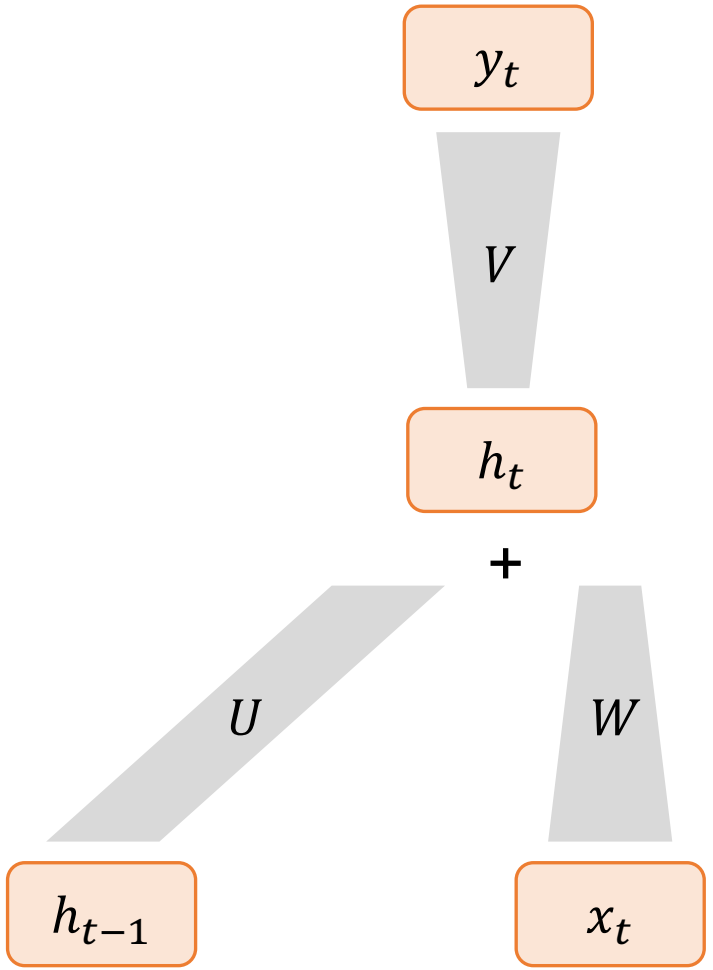
RNN

- The equation on step t is:

$$y_t = g(Vh_t)$$

$$h_t = f(Wx_t + Uh_{t-1})$$

, where f and g are activation functions.



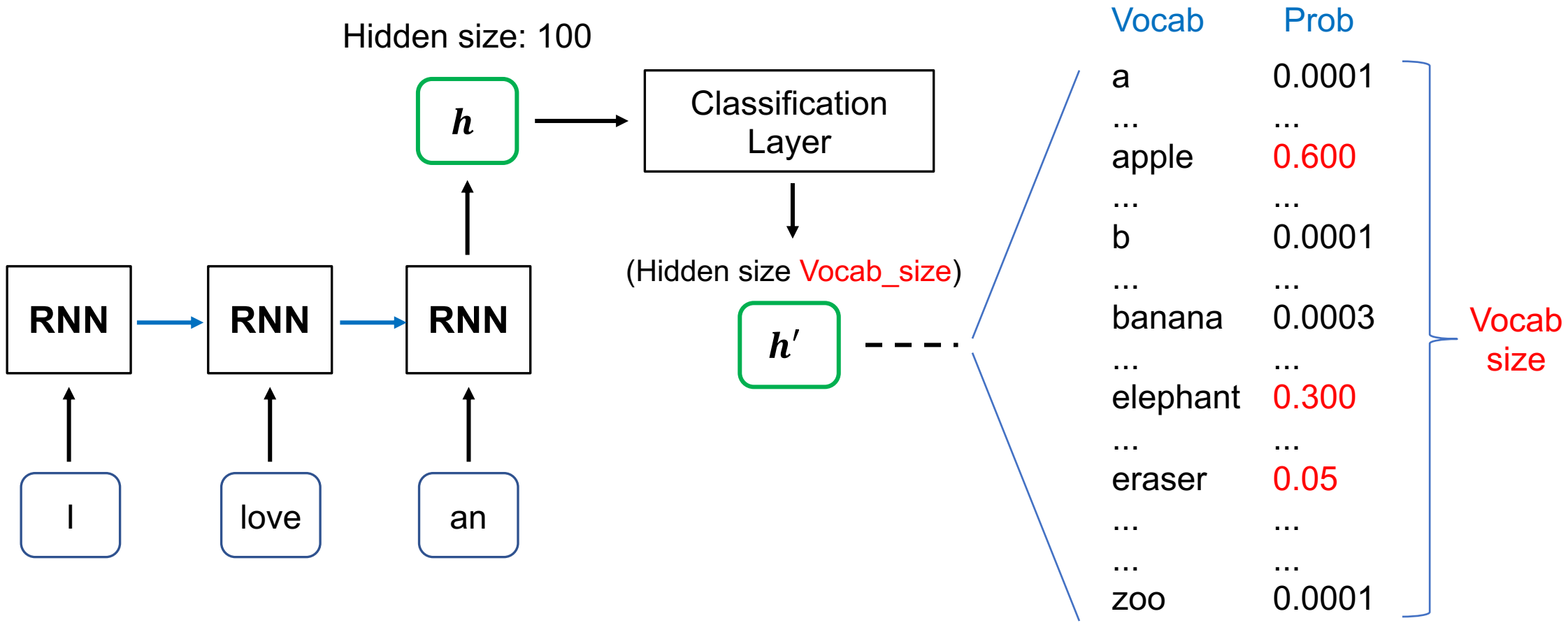
為什麼我們不能用 MLP 來做 NLP？

- Lack of Sequence Modeling:
  - In NLP tasks, understanding the sequence of words and their dependencies is crucial for accurate predictions.
- Fixed Input Size:
  - FFNs require fixed-size inputs, which can be problematic for NLP tasks where input sequences vary in length.
- Limited Contextual Information:
  - Many NLP tasks benefit from capturing long-range dependencies and understanding the broader context of a text, which is better addressed by models capable of modeling sequential data effectively.

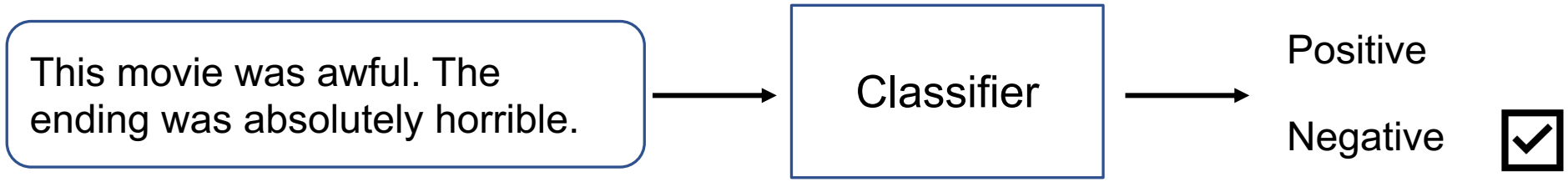
Properties of RNNs

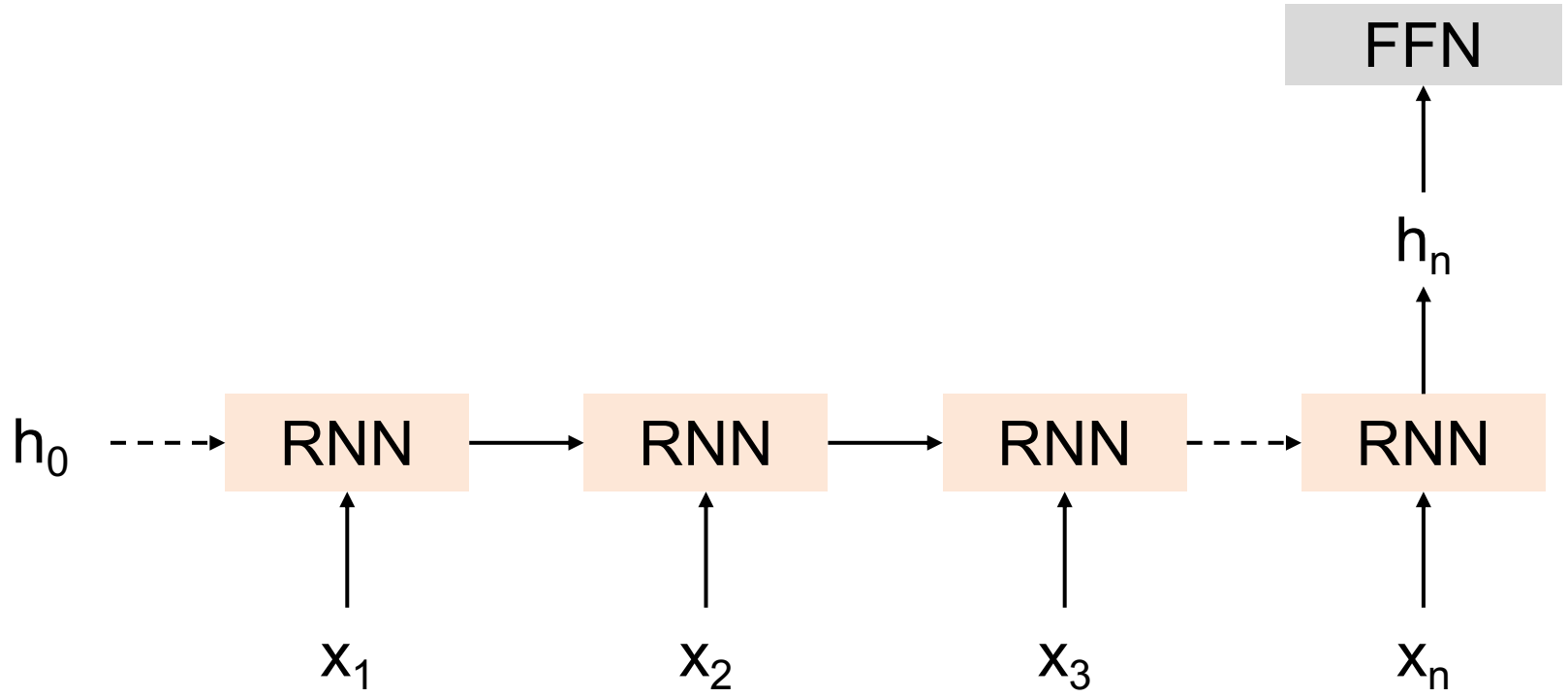
- **Sequential Processing:**
  - RNNs handle sequences, allowing them to model temporal dependencies in data.
- **Recurrent Connections:**
  - RNNs maintain internal memory, facilitating the capture of long-term dependencies.
- **Parameter Sharing:**
  - RNNs share parameters across time steps, enhancing efficiency in learning sequential data.
- **Vanishing Gradient Problem:**
  - Traditional RNNs may face vanishing gradient issues, hindering learning of long-term dependencies.

Text Generation with RNN



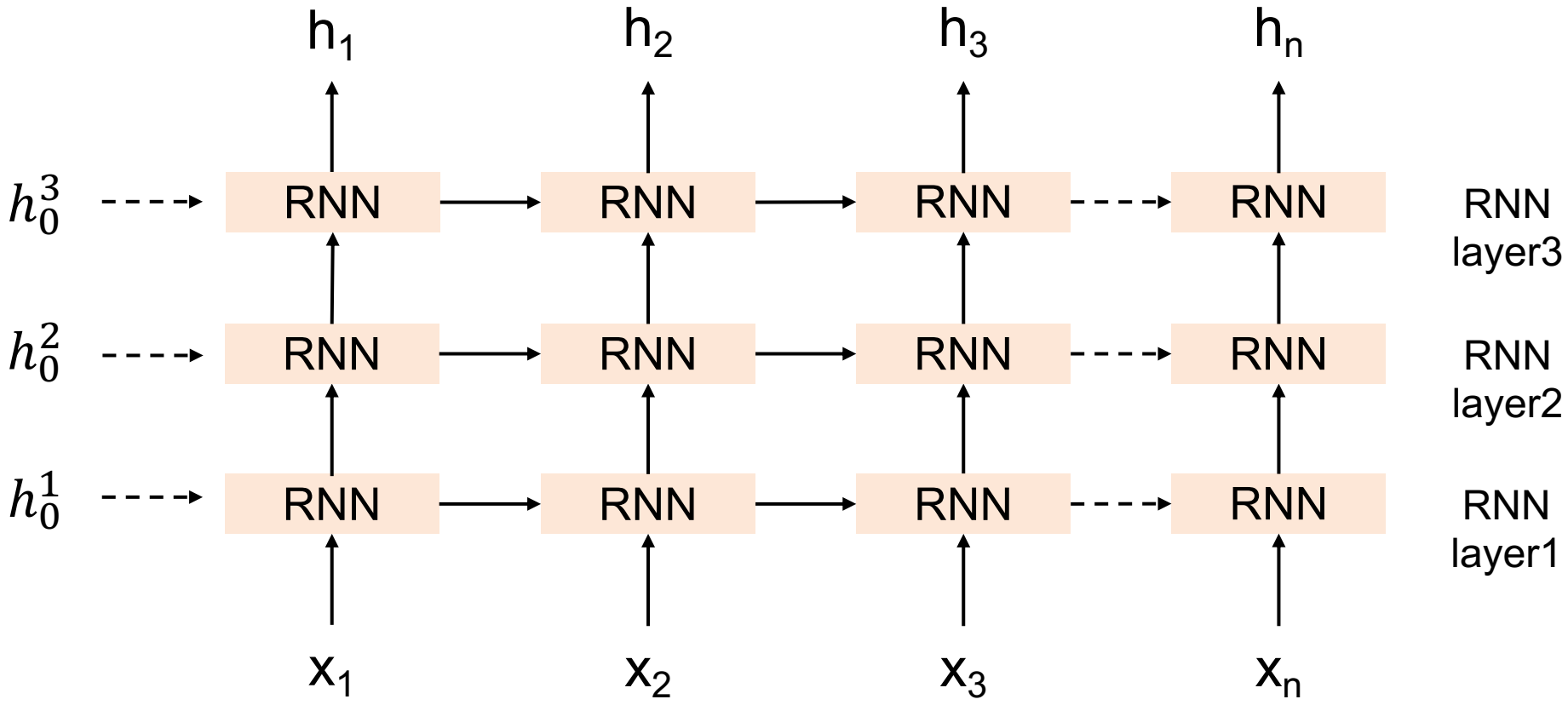
RNNs for Sequence Classification



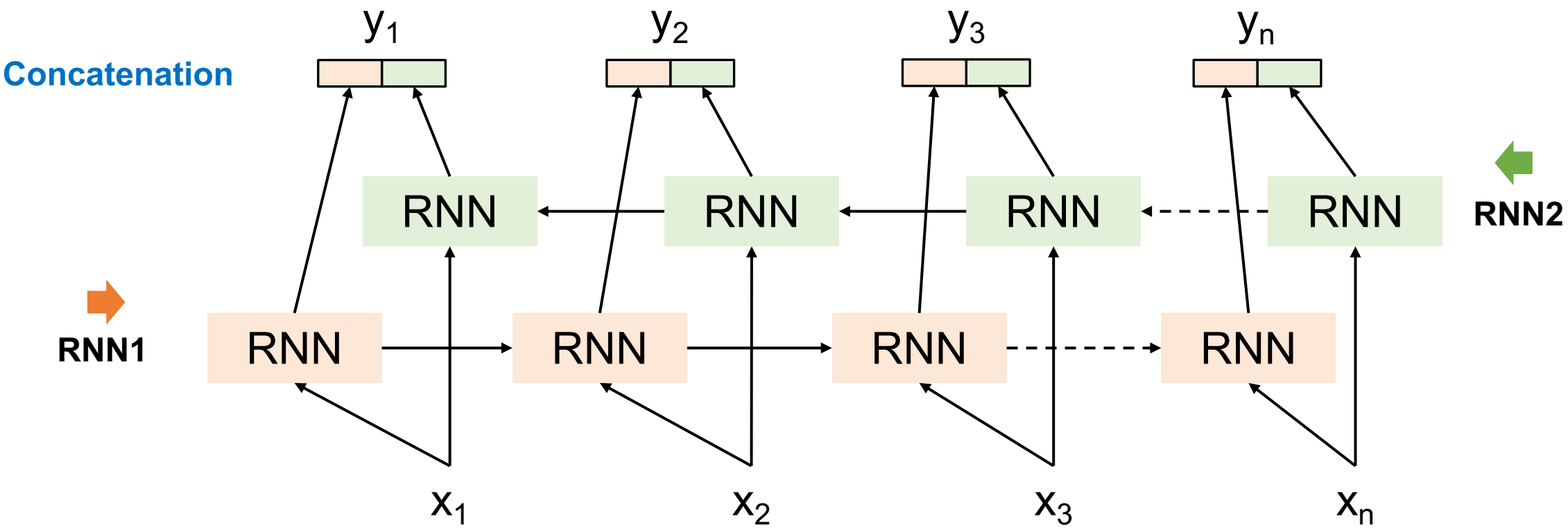


Stacked RNNs

- Stacked RNNs consist of multiple networks where the output of one layer serves as the input to a subsequent layer

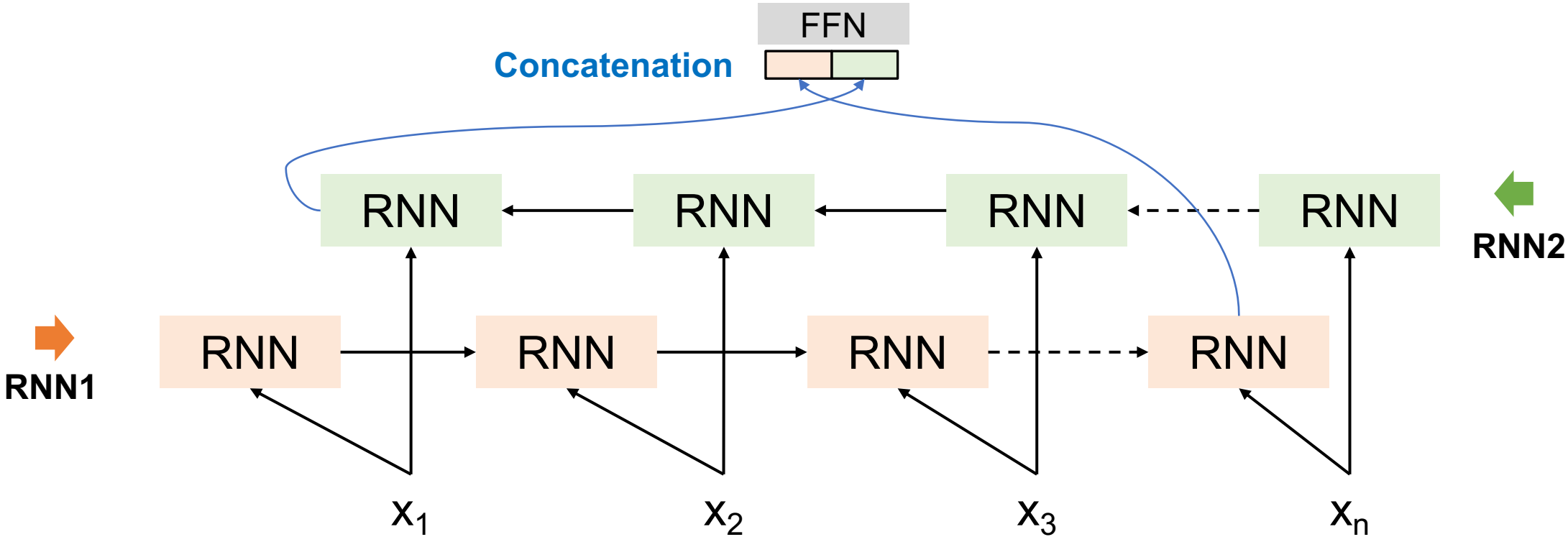


Bidirectional RNNs



Bidirectional RNNs for Sequence Classification

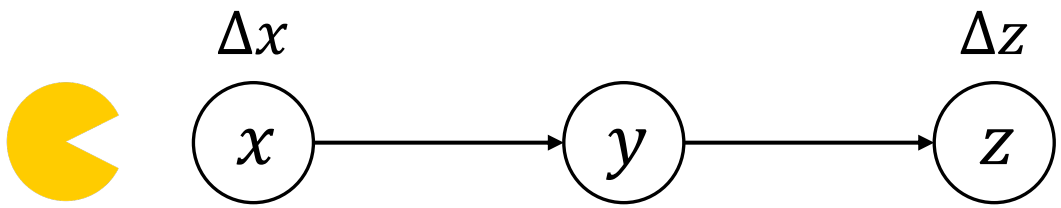
- The final hidden units from the forward and backward passes are combined to represent the entire sequence.
- This combined representation serves as input to the subsequent classifier.



Shortage of standard RNNs

- Gradient vanishing (梯度消失) problem
- Long-term dependencies cannot be handled.
  - RNNs can only remember local relationships.
  - This is result from back-propagation through-time (BPTT) during training.

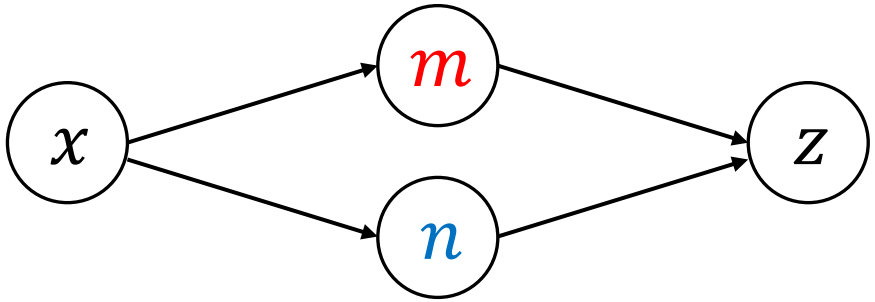
(Calculus) Chain Rule - 1



$$\frac{dz}{dx} = \frac{dz}{dy}\frac{dy}{dx}$$

- $\frac{dz}{dx}$ 代表 x 對 z 的影響，但 x 的變化會影響到 y ，接著 y 影響到 z

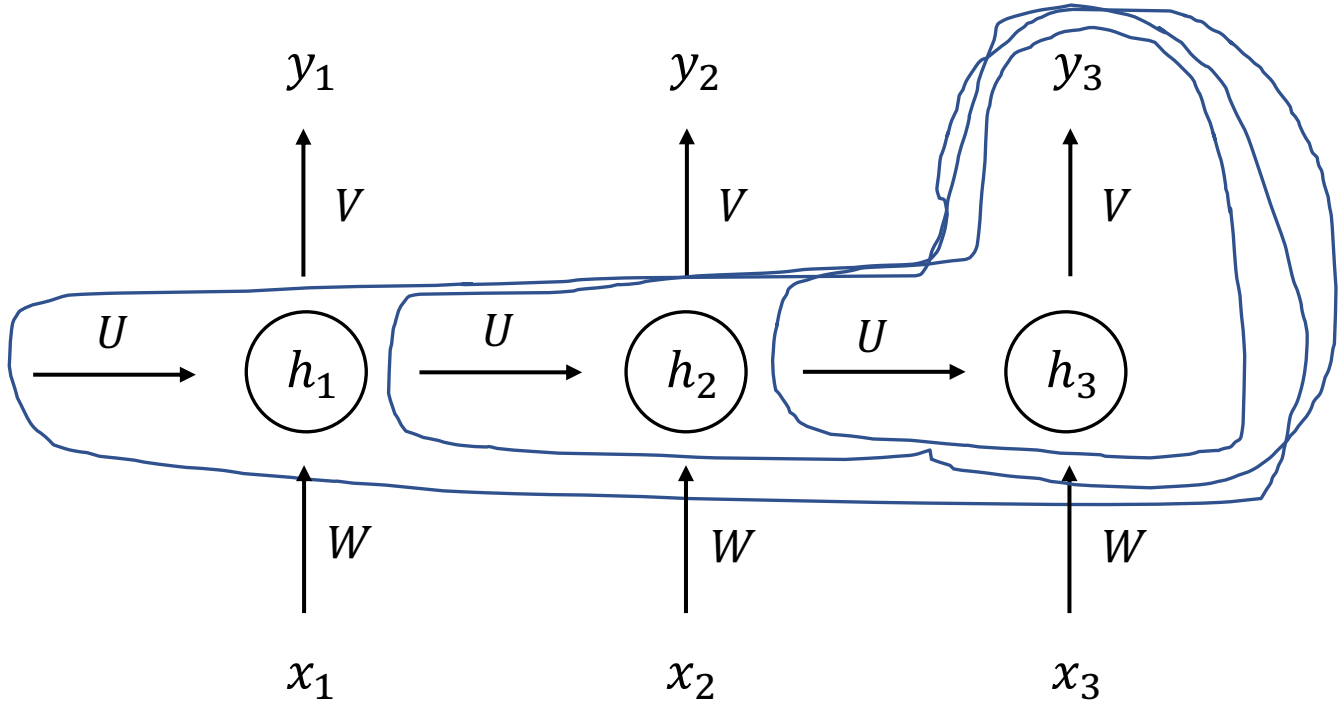
(Calculus) Chain Rule - 2

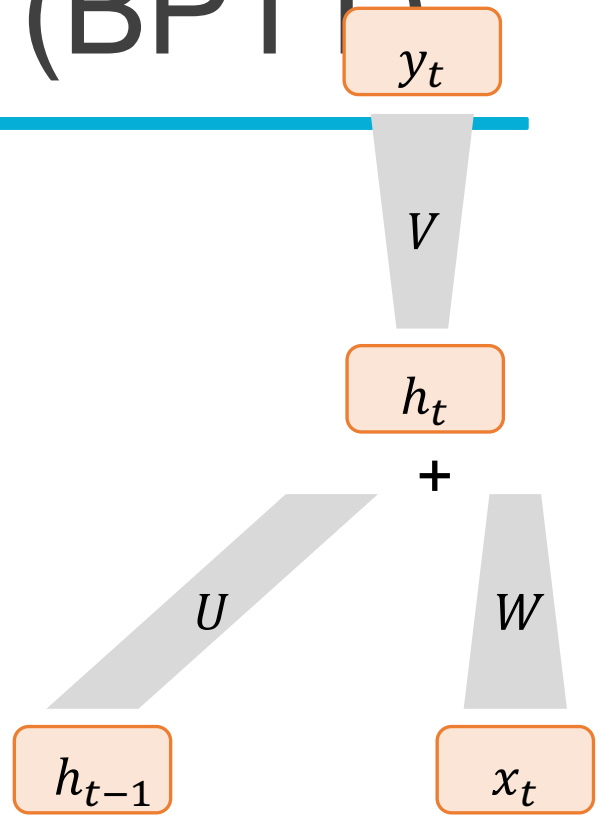


$$\frac{dz}{dx} = \frac{dz}{dm}\frac{dm}{dx} + \frac{dz}{dn}\frac{dn}{dx}$$

- $\frac{dz}{dx}$ 代表 x 對 z 的影響
  - 但 z 的變化由 m 和 n 共同影響

Back-propagation through-time (BPTT)





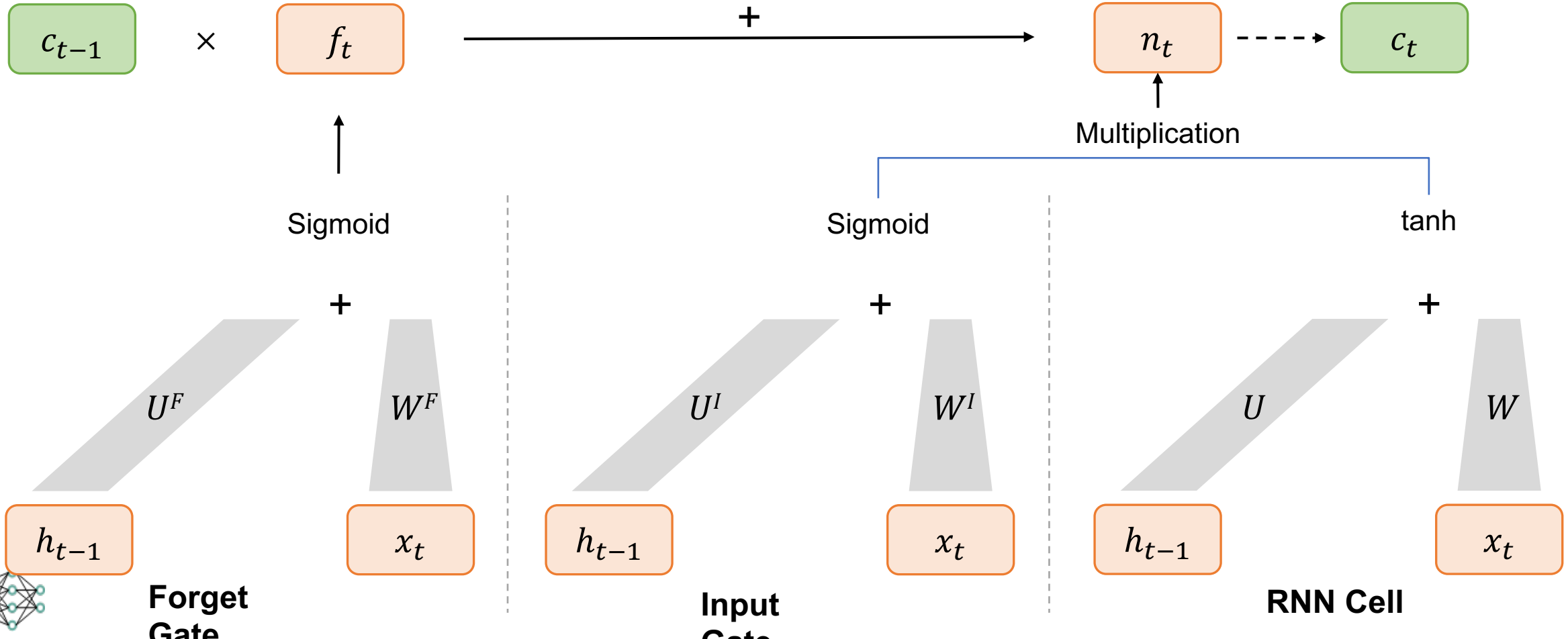
$$\frac{\partial \mathcal{L}}{\partial U} = \frac{\partial \mathcal{L}}{\partial y_3} \cdot \frac{\partial y_3}{\partial h_3} \cdot \frac{\partial h_3}{\partial U} + \frac{\partial \mathcal{L}}{\partial y_3} \cdot \frac{\partial y_3}{\partial h_3} \cdot \frac{\partial h_3}{\partial h_2} \cdot \frac{\partial h_2}{\partial U} + \underbrace{\frac{\partial \mathcal{L}}{\partial y_3} \cdot \frac{\partial y_3}{\partial h_3} \cdot \frac{\partial h_3}{\partial h_2} \cdot \frac{\partial h_2}{\partial h_1} \cdot \frac{\partial h_1}{\partial U}}_{\text{(small)}}$$

Hochreiter, Sepp, and Jürgen Schmidhuber. "Long short-term memory." Neural computation 9.8 (1997): 1735-1780.

LSTM (Long Short-Term Memory)

$$c_t = c_t \times f_t + n_t$$

- LSTM uses 4 times of weights as a standard RNN.

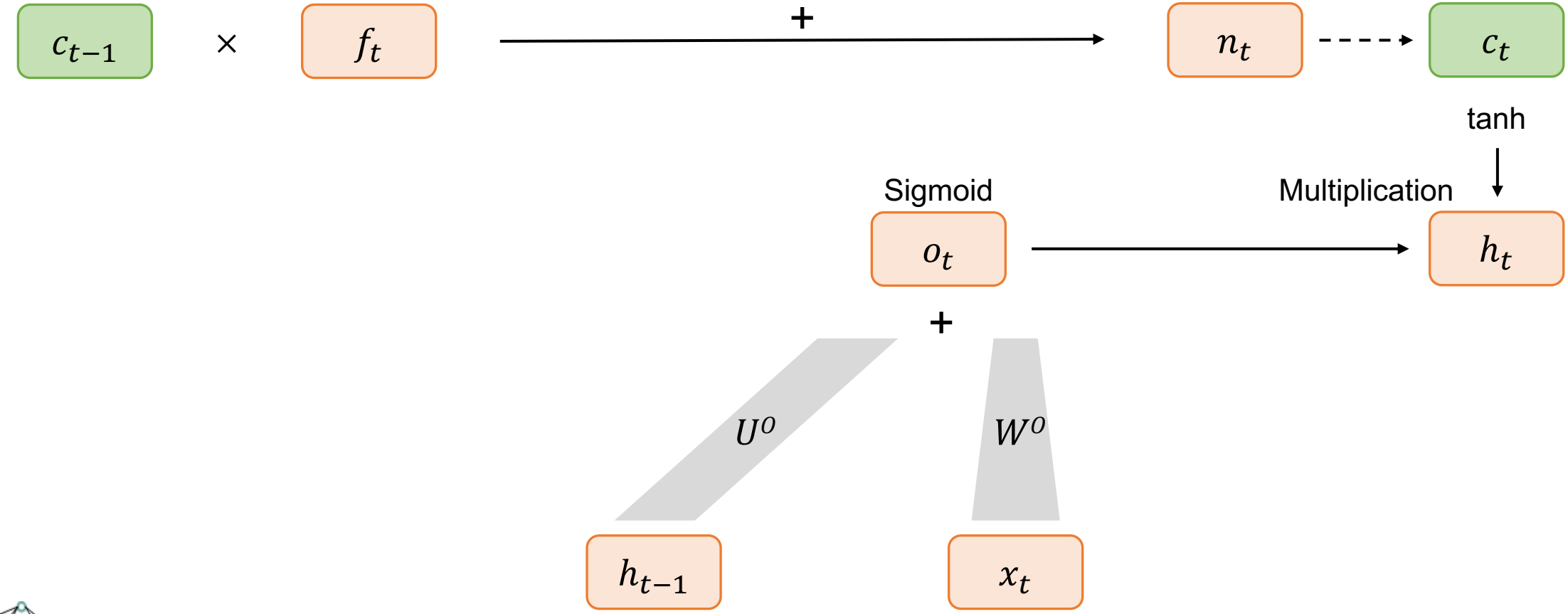


Hochreiter, Sepp, and Jürgen Schmidhuber. "Long short-term memory." Neural computation 9.8 (1997): 1735-1780.

LSTM (Long Short-Term Memory)

$$c_t = c_t \times f_t + n_t$$

- LSTM uses 4 times of weights as a standard RNN.



Output Gate

RNN vs. LSTM

	Memory Mechanism	Handles Long Dependencies?	Gradient Vanishing?
RNN	Hidden state h_t	Worse	Yes
LSTM	Memory state c_t + Hidden state h_t	Better	Much less than a standard RNN

Activation functions (啟動函數)

	值的範圍	使用情境
softmax	[0, 1] (sum up to 1)	需要讓多個值產生相加為一的機率值的時候，常用於建立分類任務模型時計算 cross-entropy loss
sigmoid	[0, 1]	需要讓單一值產生機率值的時候，常用於建立二元分類任務模型時計算 binary cross-entropy loss
tanh	[-1, 1]	常用於LSTM等RNN模型中，主要為引入非線性轉換。 tanh的值是對稱的，以零為中心，在有些訓練條件下和sigmoid相比較不會梯度消失
ReLU	[0, 正值]	常用於CNN或MLP等模型中，主要為引入非線性轉換。 計算簡單，在許多任務收斂快

Code Llama

https://arxiv.org/abs/2308.12950

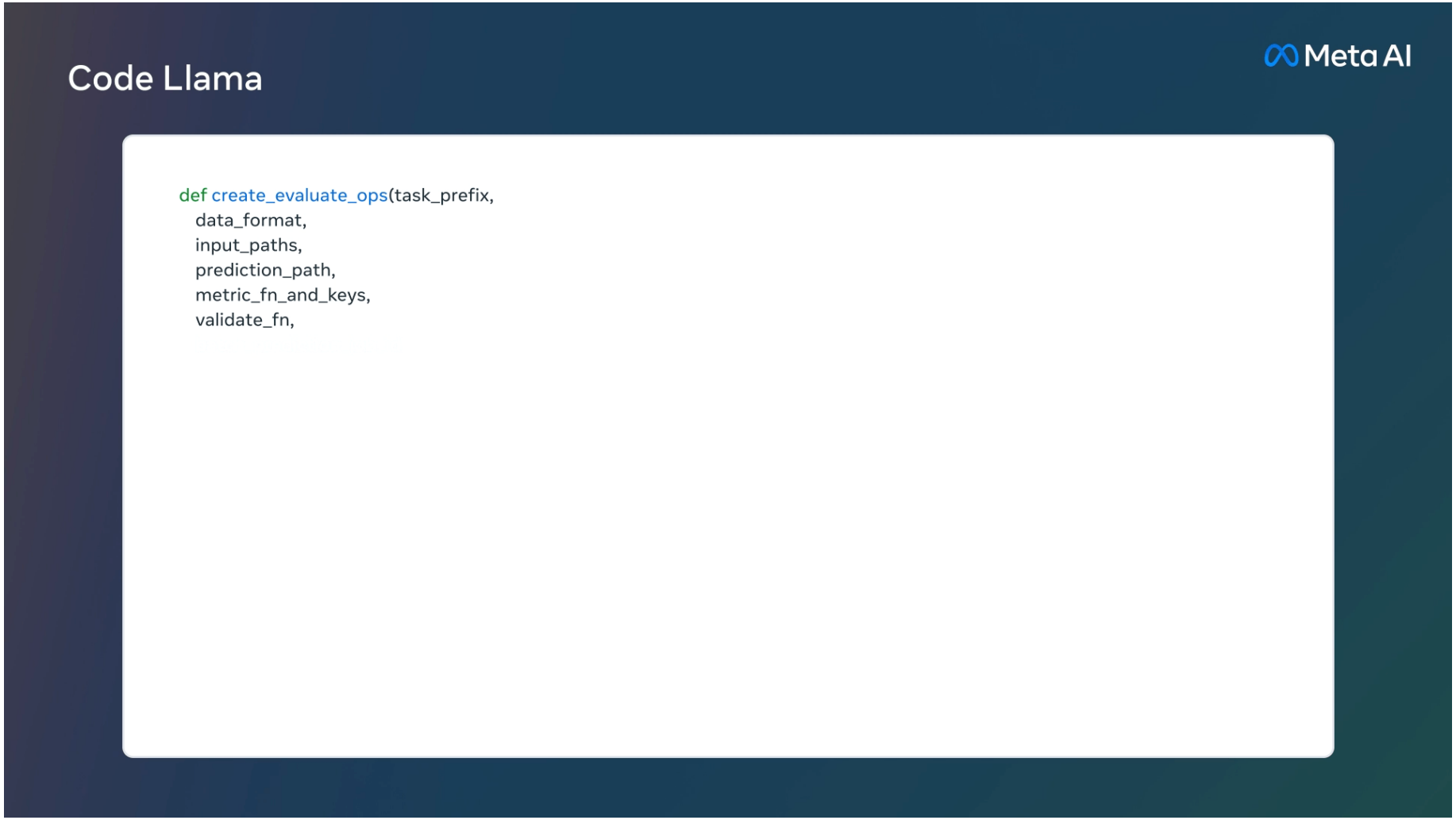
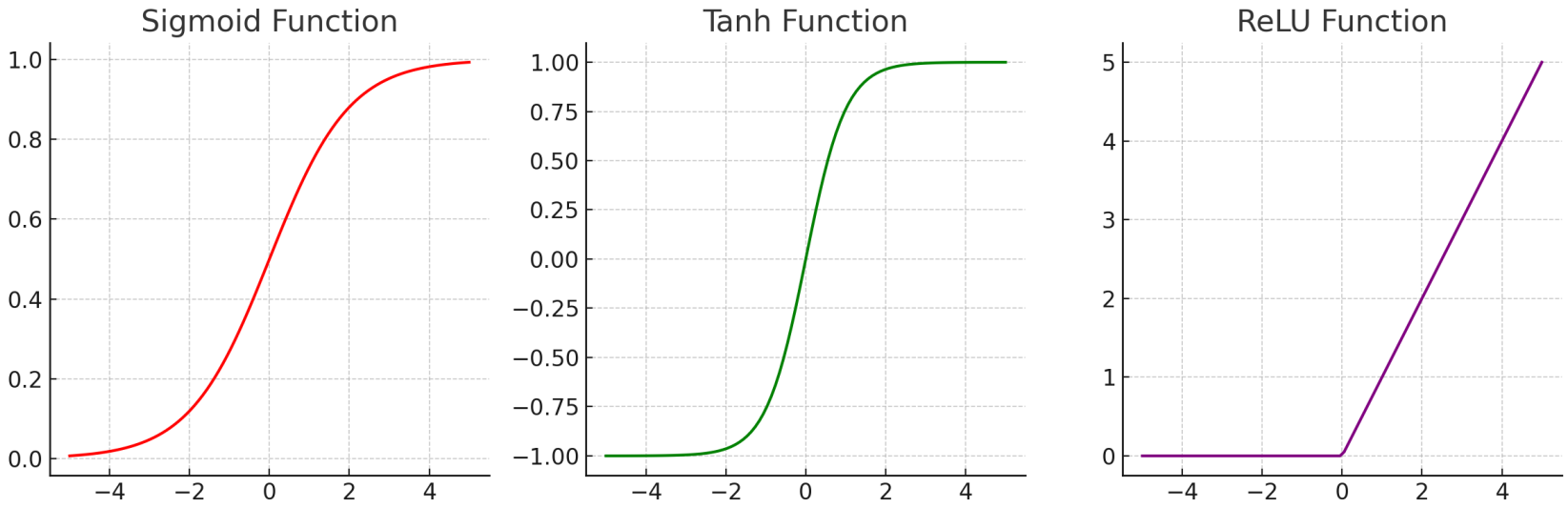


Figure source: https://ai.meta.com/blog/code-llama-large-language-model-coding/

Activation functions



Softmax:
https://datascience.stackexchange.com/questions/57005/why-there-is-no-exact-picture-of-softmax-activation-function

Thank you!

Instructor: 林英嘉

✉ yjlin@cgu.edu.tw

TA: 劉美辰

✉ m1461014@cgu.edu.tw